AI Assignment 3 - Optimize!!

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Due 4/15/26 (Wednesday)
start 4/1/26 - submitted 4/10/26

GitHub Repository: https://github.com/AshtonBeresford/ai_assignment3_optimize

Assignment Details

of 2

Optimize!

4/15/2026

10 Points Possible

Attempt 1

In Progress

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Unlimited Attempts Allowed

Details

Overview

For this assignment, you're going to solve a problem with three different methods:

1. Brute Force
2. Mixed-integer programming
3. Genetic algorithm

Problem

ONU has planned to subdivide the RiteAid into several individual businesses, and have opened an auction for proposed businesses to bid on renting the space. Below are the current bidders:

Bids for rental at former RiteAid location

Business Name	Space requested (ft^3)	Bid ($/$ft^3$)	Category
Polar Brew Coffee	1,200	4.5	Food & Beverage
Tech Haven	2,500	6.2	Electronics
Green Leaf Market	1,800	5.1	Grocery
Campus Books & Co.	2,200	4.8	Retail
FitZone Gym	3,500	3.9	Fitness
Byte Repair	900	6.8	Electronics
Urban Threads	1,600	5.5	Clothing
Clyde's Photo, Ada	1,400	4.7	Arts
GameSphere	2,000	5.9	Entertainment
Serenity Spa	2,800	4.2	Wellness
QuickPrint Center	1,100	5	Services
EcoHome Goods	1,900	5.4	Retail

Figure 1: Assignment details

Brute Force

Code in GitHub Repository in file named: brute_force.py

This code found an option that used Byte Repair, EcoHome Goods, and VR Arena as seen in figure 2 below. This results in all 6000 feet being used for a profit of $35,580 by ONU. The

brute force method averaged out to take .5 seconds to run.

```
(optimize_env) PS C:\Code_general\school\ai_assignment3_optimize\ai_assignment3_optimize> uv run brute_force.py

best path:  [0 0 0 0 0 1 0 0 0 0 0 1 0 1 0]
Space:  6000.0
Profit:  35580.0
Time to run:  0.512230799999088
```

Figure 2: Brute Force Results

Figure 3: work on Mixed-Integer Programming problem

Bids for rental at former RiteAid location

Business Name	Space requested (ft3)	Bid ($/ft3)	Category	Total Bid (Bid x Space)
Polar Brew Coffee	1,200	4.5	Food & Beverage	$5,400
Tech Haven	2,500	6.2	Electronics	$15,500
Green Leaf Market	1,800	5.1	Grocery	$9,180
Campus Books & Co.	2,200	4.8	Retail	$10,560
FitZone Gym	3,500	3.9	Fitness	$13,650
Byte Repair	900	6.8	Electronics	$6,120
Urban Threads	1,600	5.5	Clothing	$8,800
Clyde's Photo, Ada	1,400	4.7	Arts	$6,580
GameSphere	2,000	5.9	Entertainment	$11,800
Serenity Spa	2,800	4.2	Wellness	$11,760
QuickPrint Center	1,100	5	Services	$5,500
EcoHome Goods	1,900	5.4	Retail	$10,260
Smoothie Spot	1,000	4.6	Food & Beverage	$4,600
VR Arena	3,200	6	Entertainment	$19,200
Study Lounge Café	2,100	4.9	Food & Beverage	$10,290

Figure 4: Table with Total Bid section

This code found an option that used Byte Repair, EcoHome Goods, and VR Arena as seen in figure 5 below. This results in all 6000 feet being used for a profit of $35,580 by ONU. The mixed-integer algorithm averaged out to taking .00065 seconds per run.

```
(optimize_env) PS C:\Code_general\school\ai_assignment3_optimize\ai_assignment3_optimize> uv run mixed_int.py
        message: Optimization terminated successfully. (HiGHS Status 7: Optimal)
        success: True
         status: 0
            fun: -35580.0
              x: [ 0.000e+00  0.000e+00 ...  1.000e+00  0.000e+00]
  mip_node_count: 1
  mip_dual_bound: -35580.0
        mip_gap: 0.0

  [0. 0. 0. 0. 0. 1. 0. 0. 0. 0. 0. 1. 0. 1. 0.]
Byte Repair  - Space Requested:  900.0 , Bid:  6.8 , Total Bid:  6120.0
EcoHome Goods  - Space Requested:  1900.0 , Bid:  5.4 , Total Bid:  10260.0
VR Arena  - Space Requested:  3200.0 , Bid:  6.0 , Total Bid:  19200.0

Total space used =  6000.0  ft^3. Which leaves  0.0 ft^3 unused.
Total profit for ONU = $ 35580.0 .
Time to run:  0.0006410999922081828
```

Figure 5: Mixed Integer results

This code found would bounce around based on population size and number of runs, however I tried to settle on a reasonable amount of generations (40) and population size (20) which results with: Byte Repair, Urban Threads, Gamesphere, and Quickprint Center as seen in figure 6 below. This results in 5600 feet being used, leaving 400 feet unused, for a profit of $32,220 by ONU. The Genetic Algorithm averaged out to taking .85 seconds to run.

```
(optimize_env) PS C:\Code_general\school\ai_assignment3_optimize\ai_assignment3_optimize> uv run genetic.py
Best solution found: [0 0 0 0 0 1 1 0 1 0 1 0 0 0 0]
Function value: [-32220.]
Constraint violation: [0.]
Byte Repair  - Space Requested:  900 , Bid:  6.8 , Total Bid:  6120
Urban Threads  - Space Requested:  1600 , Bid:  5.5 , Total Bid:  8800
GameSphere  - Space Requested:  2000 , Bid:  5.9 , Total Bid:  11800
QuickPrint Center  - Space Requested:  1100 , Bid:  5 , Total Bid:  5500

Total space used =  5600  ft^3. Which leaves  400 ft^3 unused.
Total profit for ONU = $ 32220 .
Time to run:  0.8565121000283398
```

Figure 6: Genetic Algorithm Results

Surprisingly the brute force method didn't take the longest to run on average, as that achievement would be taken by the genetic algorithm. In order of from fastest to slowest the order is: mixed-integer, brute force, genetic. However, I believe that in a scaled up version of this, for example, if we had 100x the space to work with and 5,000 more choices of businesses, that brute force would scale the worst and be take the most resources to run. The genetic algorithm suffered from all of the internal setup of the algorithm itself which causes it to run longer than expected, I also could have included some lines of code unnecessarily by not knowing what specific lines or spots would cause the program to run and finish. I believe that in larger scales the genetic algorithm would perform better as this set up time would become negligible with a large enough problem.

The genetic algorithm therefore ends up as the slowest, in this example, and the only algorithm without the perfect solution. While this may imply that this algorithm is the worst, there are very good uses for it. As mentioned above, this algorithm does better than algorithms like brute force when scaling. Another benefit of this type of algorithm is that its easier to define the problem and set it up since you don't need to know exactly how it works and the answer it comes out to. While it will almost never provide a perfectly optimized solution, it will, if set up correctly, always provide an optimal solution, roughly in the top 80th percentile or so. The algorithm can be set up easier than mixed-integer problems when their multiple objectives, as the genetic algorithm itself figures out how to way the different objectives, while you would have to do that yourself with mixed integer (or at least figure out how they interact with each other). This means that it can be easier to set up multi dimensional problems, more complicated problems than a mixed-integer algorithm while scaling better than the brute force algorithm, therefore providing worth to genetic algorithms in situations like this.

So to reiterate, the truly optimal solution, as founded by the brute force and mixed-integer programing methods would be: Accepting the bids from Byte Repair, EcoHome Goods, and VR Arena. This outcome utilizes all 6000 feet and results in a total profit of $35,580 by ONU.